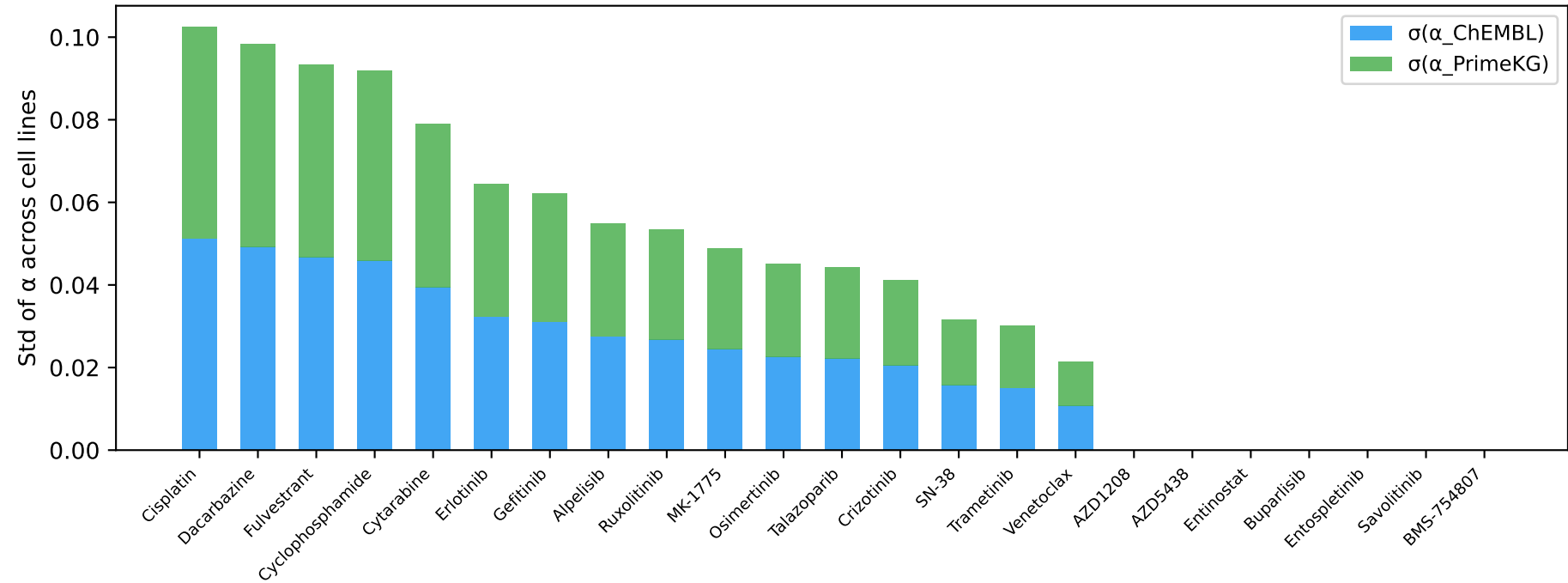
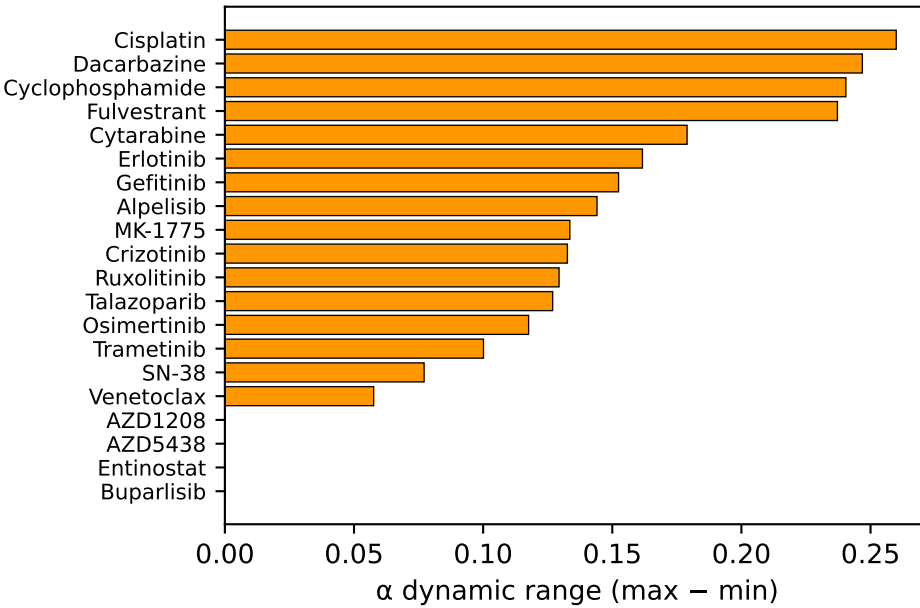


Scenario B — B02: Dynamic KG Attention: Static Prior Networks  
Become Context-Dependent in the World Model

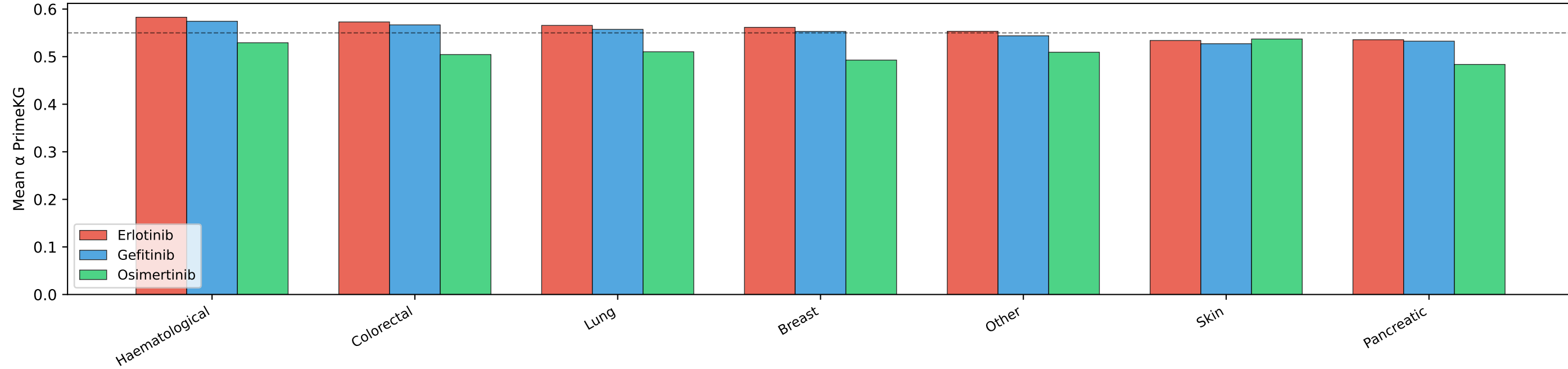
A. Per-Drug Attention Variance (Static→Dynamic Transformation)  
Higher  $\sigma$  = KG attention is more context-dependent



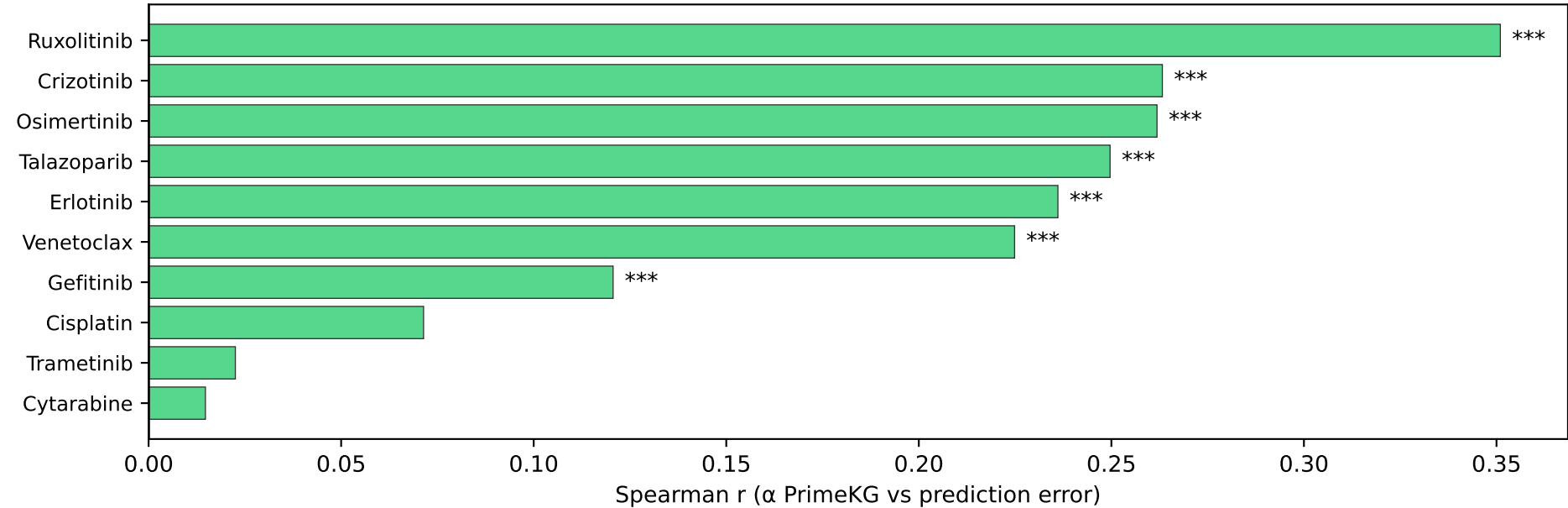
B. Alpha Dynamic Range  
(per drug across cells)



C. EGFR Inhibitors: Dynamic PrimeKG Attention by Cancer Type  
(protein interaction network weighted differently per cell context)



D. Higher PrimeKG Attention → Lower Error  
(negative  $r$  = more PrimeKG attention → better prediction)



E. Summary

Static → Dynamic KG:  
DRKG: 0% test drug coverage  
→ auto-suppressed ( $\alpha \approx 0$ )  
  
Alpha varies per cell context:  
Max  $\sigma(\alpha)$ : 0.0512  
Max range: 0.2599  
  
Erlotinib (EGFR inhibitor):  
Lung  $\alpha_{\text{PrimeKG}}$ : 0.5657  
Other  $\alpha_{\text{PrimeKG}}$ : 0.5580  
MW  $p=0.011$   
  
→ Same drug, different KG  
weighting by cancer context